21 Maret 2022 / Senin.

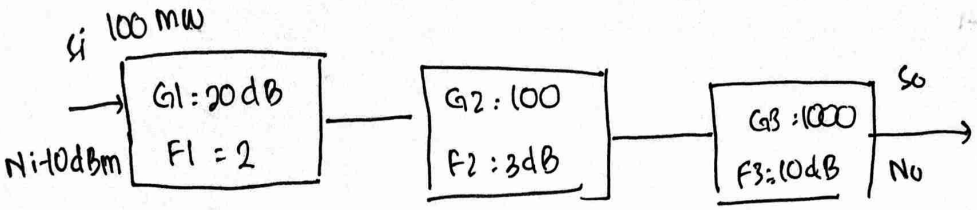
UTS SISKOM 1.

Semester 4 / 2021-2022

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Kelas : 2NK

Jawablah soal-soal dibawah ini!

1. A. Penguat bertingkat



Hitunglah

a). S/N di input (boleh dalam rasio ataupun dBm).

$Ni = -10\,dBm \rightarrow 10^{-10/10} = 0{,}1\ mw$

$Si = 100\ mw$

$\frac{Si}{Ni} = \frac{100\ mw}{0{,}1\ mw} = 1000$ (rasio)

$G2 = 100\ mw / 20\,dB$

$F2 = 3dB \rightarrow 10^{3/10} = 1.99$

$F3 = 10dB \rightarrow 10\ mw.$

$G1 = 20dB = 100mw$

$\left[\frac{Si}{Ni}\right] dB = (10 \log (Si/Ni)$ rasio.

$= 10 \log 1000$

$= 30\,dB.$ (dB)

b.) Noise Figure Total (boleh dalam rasio ataupun dB.

$Ft = F1 + \left(\frac{F2-1}{G1}\right) + \left(\frac{F2-1}{G1 \times G2}\right)$ (dalam rasio).

$= 2 + \left(\frac{1.99-1}{100}\right) + \left(\frac{10-1}{100 \times 100}\right)$

$= 2 + \left(\frac{0.99}{100}\right) + \left(\frac{9}{10^4}\right)$

$= 3 + (0.0099) + (0{,}0005)$

$= 3{,}0108$ //

(Ft dalam dB)

$Ft = 10 \log (Ft)$ rasio

$= 10 \log 3{,}0108$

$= 4{,}78\ dB.$

1MHz 1,05MHz.

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$G_t = G_1 \times G_2 \times G_3$
$= 100 \times 100 \times 1000$
$= 10^7$

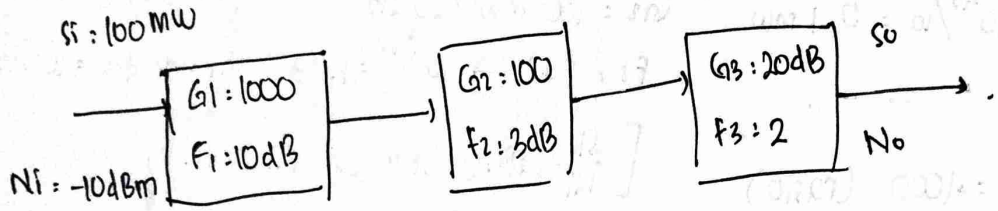
c. S/N di output (boleh dalam rasio/dB).

$S_o = G_t \text{ (rasio)} \times S_i \text{ (rasio)}$
$= 10^7 \times 100 \text{ mw}$
$= 10^9 \text{ mw}$

$N_o = G_t \text{ (rasio)} \times N_i \text{ (rasio)}$
$= 10^7 \times 10^{-1}$
$= 10^6 \text{ mw} //$

$\frac{S_o}{N_o} = \frac{10^9}{10^6} = 10^3 : 1000 \text{ mw} //$

2. Penguat bertingkat gambar 2 berikut.



Hitung :

d.) S/N di input.

$N_i = -10 \text{ dBm} \rightarrow 10^{-10/10} : 0,1 \text{ mw}$
$S_i : 100 \text{ mw}$

$\frac{S_i}{N_i} = \frac{100 \text{ mw}}{0,1 \text{ mw}} : 1000 \text{ (rasio)}$

$F_2 = 10^{3/10} : 1,99 \text{ mw}$
$G_2 : 100 \text{ mw}$

$F_1 : 10\text{dB} \rightarrow 10^{10/10} = 10 \text{ mw}$
$F_3 = 2\text{dB} \rightarrow 10^{2/10} = 1,58$

$\left[\frac{S_i}{N_i}\right] \text{dB} : 10 \log 1000$
$= 30 \text{ dB}$ (dB).

e.) Noise Ft (boleh dalam rasio/dB).

$F_t = F_1 + \left(\frac{F_2 - 1}{G_1}\right) + \left(\frac{F_3 - 1}{G_1 \times G_2}\right)$

$= 10 + \left(\frac{1.99 - 1}{1000}\right) + \left(\frac{2 - 1}{1000 \times 100}\right)$

$= 10 + \left(\frac{0,99}{10^3}\right) + \left(\frac{1}{10^5}\right)$

$= 10 + (9,9 \times 10^{-4}) + (1 \times 10^{-5})$
$= 10,00100053. \text{ (rasio.)}$

(dalam dB)

: 10 log (10,00100 53)

: 10,0004365 dB. //

f.) S/N di output (rasio / dB).

So : Gt (rasio) x Si (rasio)
: 10^7 x 1000
: 10^{10} //

No : Gt (rasio) x Ni (rasio)
: 10^7 x 10^{-1}
: 10^6 //

Gt : G1 x G2 x G3
: 1000 x 100 x 100
: 10^7

$\frac{So}{No} = \frac{Si}{Ni} - F$
= 30dB - 10,0043
= 19.99
= 20 dB.

$\frac{So}{No} = \frac{10^{10}}{10^6} = 10^4$ // : 10.000 mw

g.) dari hasil perhitungan berdasarkan pada gambar 1 dan 2, susunan penguat yang. yang lebih baik. adalah. gambar. 1 karena pada gambar 1 noise yang lebih kecil dibandingkan. gambar 2. Lalu nilai S/N di output pada gambar 1 memiliki nilai yang lebih besar dibandingkan. nilai S/N di output gambar 2 //.

2. Dik : Modulator AM - DSB-FC.
Daya pancar total = 10 kw (Pt).
Teg. maks (VAM(t)) : 20 V
Vm(t) : 10 sin 2π 5000 volt.
frek sinyal pembawa : 100 KHz.

Jawablah.
a.) Besar indeks modulasi m

Pt : 10 kW
Vm(t) : 10 sin 2π 5.000 volt.
t = 100 KHz.
A(t) = 20 V.

A(t) : Vcmax + Vm(t)
20 : Vcmax + 10
Vcmax : 10
$m = \frac{Vm}{Vc} = \frac{10}{10} = 1$ //

6.) Presentase modulasi % m

$$\%m = \frac{Vm}{Vc} \times 100\%$$

$$= \frac{1}{1} \times 100\%$$

$$= 100\ \%$$

c.) Daya sinyal informasi.

$$Pm = \frac{1}{2} Pc$$

$$= \frac{1}{2} \cdot 6{,}67\ KW$$

$$= 3{,}335\ KW$$

d.) Daya yang dibutuhkan untuk carrier.

$$Pt = Pc\left(1 + \frac{m^2}{2}\right)$$

$$10KW = PC\left(1 + \frac{1^2}{2}\right)$$

$$PC = \frac{10\ kw}{1{,}5}$$

$$PC = 6{,}67\ KW$$

e.) Besar Bandwith sinyal AM

$$Bw = 2 . Fm$$

$$= 2 . 5\ kHz$$

$$= 10\ kHz$$

f.). Besar tegangan max sinyal pembawa.

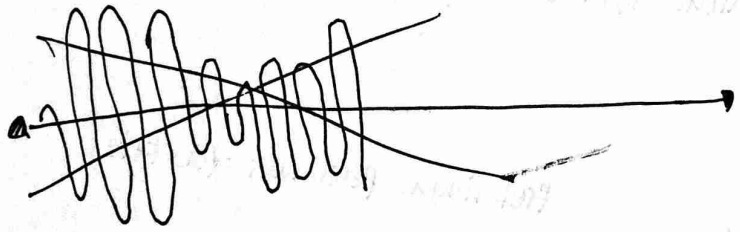
$$Vmax = Vc + Vm$$

$$20 = Vc + 10$$

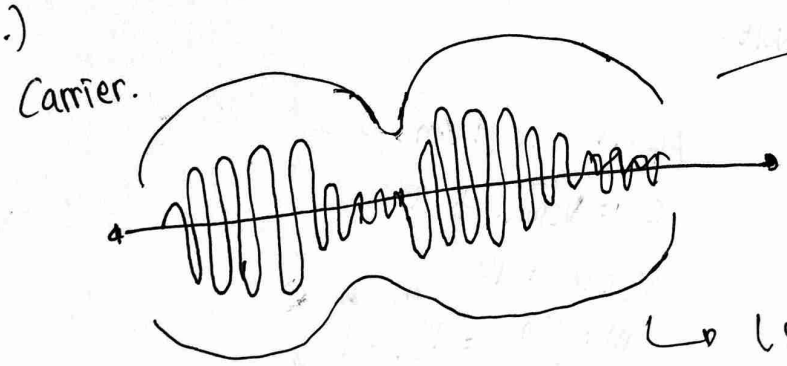
$$Vc = 20 - 10$$

$$Vc = 10\ watt.$$

9.).

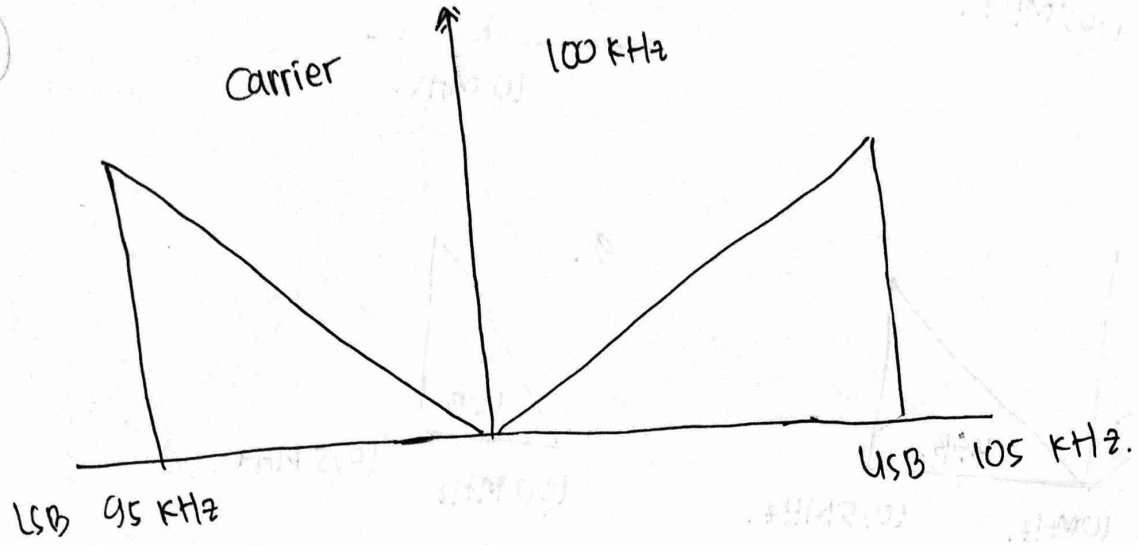


9.)

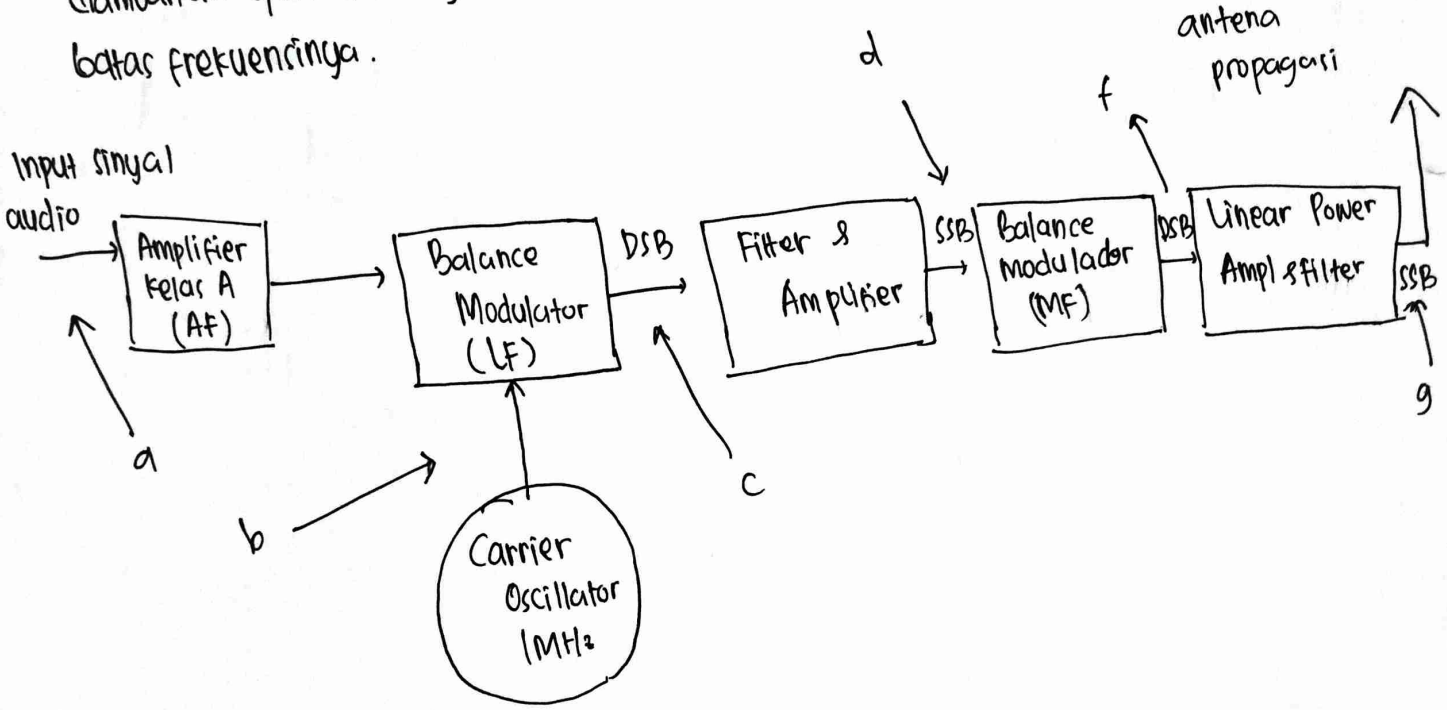


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h.

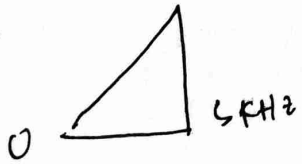


3. Dik : diagram blok pemancar SSB seperti berikut.
Gambarkan spektrum sinyal di titik a, b, c, d, e, f, dan g. dan beri batas frekuensinya.



Jawab.

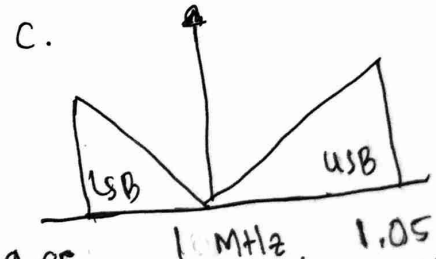
a.



b.



c.

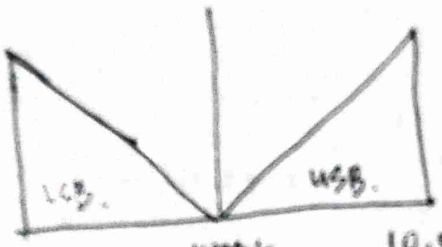


d.



1MHz 1,05MHz.

e.



10MHz.

f.

9,95 MHz 10MHz. 10,5MHz.

g.



100 MHz 10,5 MHz.

9.)